



ZenFi

FINANCIAL PEACE OF MIND

STUDY GROUP 1

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Problem Statement



Have you ever clicked the pay button and instead of the tick you see a popup saying "insufficient balance" ?



In today's world, where technology is constantly evolving, many people find it challenging to manage their money effectively, especially with the rise of online shopping. While shopping online is convenient, it can also make it harder to keep track of our spending. This can lead to spending too much money, feeling stressed about finances, and not having a good handle on our money situation.

User Persona

Demographics

- Name: Sarah
- Age: 18
- Student
- Education: ABC College
- Location: Urban area
- Tech Savvy: Moderate to high



Behaviors & Habits

- Frequently browses online stores during breaks.
- Impulsively buys items, especially during flash sales.
- Open to seeking out resources and strategies for financial management.
- Willing to adopt new habits and practices to improve financial health.
- Shows concern and motivation to overcome impulse spending tendencies.

Pain Points & Frustrations

- Impulse spending during online browsing sessions.
- Difficulty tracking where her money is going.
- Feeling of guilt or regret after making unplanned purchases.
- Lack of clarity on financial priorities and goals.

Needs & Goals

- Desire to avoid falling for tempting deals and flash sales.
- Want to prioritize financial goals and savings.
- Seek resources for improving financial literacy.
- Need strategies to curb impulse spending habits.
- Want to develop a proactive approach to managing money.

OUR GOAL IS TO HOOK YOU UP WITH:

- Automate transaction tracking for online purchases
- Categorize purchases for better organization
- Provide alerts for nearing spending limits
- Offer real-time insights into spending habits using ChatBot
- Empower users to make informed financial decisions
- Alleviate financial stress and promote financial security
- Provide an optimal buying option



Solution Overview

The solution comprises a Chrome extension designed to enhance users' financial habits through real-time tracking, transaction reminders, and ChatBot interaction. Here's an overview of how it works:

1. Real-time Savings Tracking:

- Our Chrome Extension allows users to set specific savings goals and track their progress in real time.
- Users can visualize their financial journey, motivating them to stay on track and achieve their goals.

2. Transaction Reminder Extension:

- The Chrome extension serves as a transaction reminder tool, popping up whenever users attempt a transaction online on an e-commerce website.
- If users proceed with the transaction, they have the option to record it within the app.

3. ChatBot Integration

- The main page includes a chatbot feature that interacts with users to provide detailed spending analysis.
- The chatbot retrieves data from the backend database, including the expense table, and offers personalized insights and recommendations based on users' spending patterns.

4. Gmail Integration to track records

- The webpage automatically tracks expenses by analyzing users' Gmail inboxes for e-commerce transaction-related emails.
- It identifies the order-related emails and then data is extracted from only these emails in a specified format, which is automatically added to the user's expenses.
- It is a full working model, which is flexible enough to take up different queries and give desired output.

5. Option for optimal purchases

- Users can utilize our feature to extract product information from e-commerce websites such as Amazon and Flipkart.
- This feature allows users to compare prices, find the most optimal deals, and make informed purchasing decisions.

6. Price Prediction

- Predict the best time to buy a product based on the price history of the product.

Implementation Details

1) Django Framework for Database Management:

- Utilized Django, a high-level Python web framework, to handle database management.
- Employed Django's ORM (Object-Relational Mapping) to facilitate seamless interaction with the backend database, storing user data such as expenses and transaction history.
- Leveraged Django's built-in features for user authentication, session management, and data modeling to ensure secure and efficient database operations.

2) HTML, CSS, and JavaScript for UI Development:

- Implemented the user interface using HTML for structure, CSS for styling, and JavaScript for interactivity.
- Designed intuitive and responsive UI components to enhance user experience and facilitate smooth navigation.
- Utilized JavaScript frameworks like React or Vue.js for dynamic UI elements and seamless data interaction.

3) Manifest.json for Chrome Extension:

- Created a manifest.json file to define the metadata and configuration of the Chrome extension.
- Specified details such as extension name, version, description, icons, permissions, and content scripts.
- Configure background scripts and event handlers to manage extension functionality and communication with the backend.

4) Gemini and Python for Chatbot Integration:

- Utilized Gemini, an open-source conversational AI platform, to develop the chatbot interface.
- Integrated Gemini with Python backend to handle user queries, analyze spending patterns, and provide personalized financial insights.
- Leveraged natural language processing (NLP) techniques and machine learning algorithms to improve chatbot understanding and response accuracy.

5) Python and JSON for Future Predictions:

- Implemented future prediction algorithms using Python libraries such as Pandas, NumPy, and sci-kit-learn.
- Trained predictive models based on historical spending data and user behavior patterns.
- Utilized JSON (JavaScript Object Notation) for data interchange between the backend server and frontend UI, enabling seamless integration of predictive analytics into the application.

6) Gmail Integration for automatic tracking of Expenses

- Users sign up and log in using their credentials, and their unique user ID is retrieved from cookies. The system securely accesses their Gmail inbox using the provided app password.
- Email Processing: Emails from the past day are processed, identifying order-related emails based on predefined criteria. Extracted data is formatted, stored, and added to the main expenses table.
- Gemini Integration for NLP: Gemini enhances expense tracking accuracy by processing email content using natural language processing techniques, extracting detailed information.
- Expense tracking and Updates: Total expenses for each user are updated with the extracted email expenses and displayed in a user-friendly table. Users can initiate this process by clicking the "Email Expenses" button.

7) Integration with ML ALGORITHMS:

- Developed an ML model utilizing linear regression for predicting future prices.
- Utilized timestamp conversion of dates to enable accurate predictions based on time-series data.
- Integrated the ML model into the prediction pipeline to enhance accuracy and reliability.
- Leveraged linear regression to capture complex relationships between variables, facilitating informed predictions.
- Employed feature engineering techniques to extract meaningful insights from the data, improving prediction performance.

Course References

Concepts of OOPS:

- Incorporated concepts from Object-Oriented Programming (OOP) into our implementation to improve efficiency, organization, and maintainability.

Concepts of DBMS:

- Incorporated concepts from Database Management Systems (DBMS) into our implementation to improve efficiency.
- Implemented relationships between different tables using foreign keys to establish connections between user expenses and user statuses, enabling seamless querying and analysis of user financial data.

Concepts of DSA:

- Algorithms are used to calculate the total expenses and the financial score of the user.

Concepts of Problem Solving and Design Thinking:

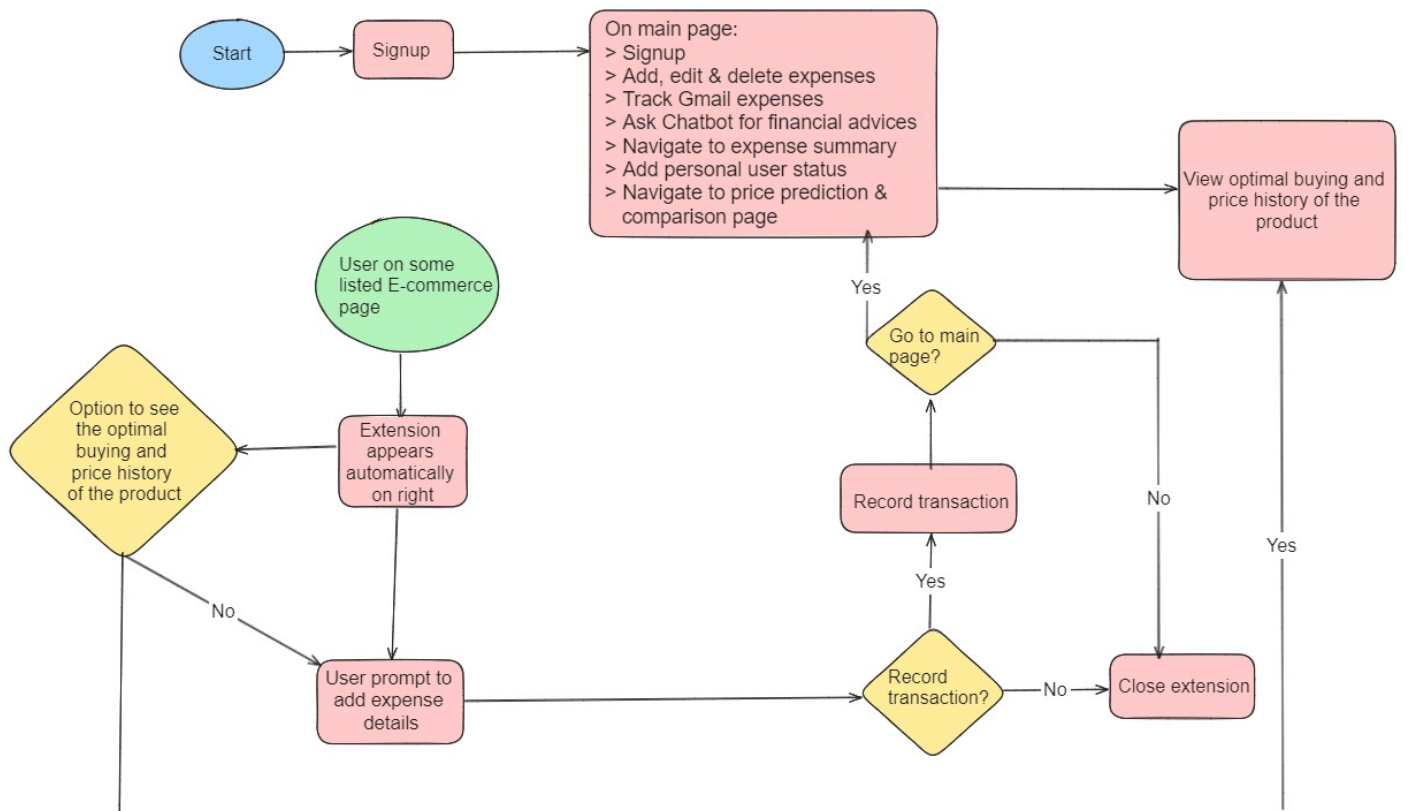
- The first step in design thinking is to understand the users' needs and pain points. In the case of an expense tracker, this involves identifying the challenges users face in managing their expenses, budgeting, or analyzing financial patterns.



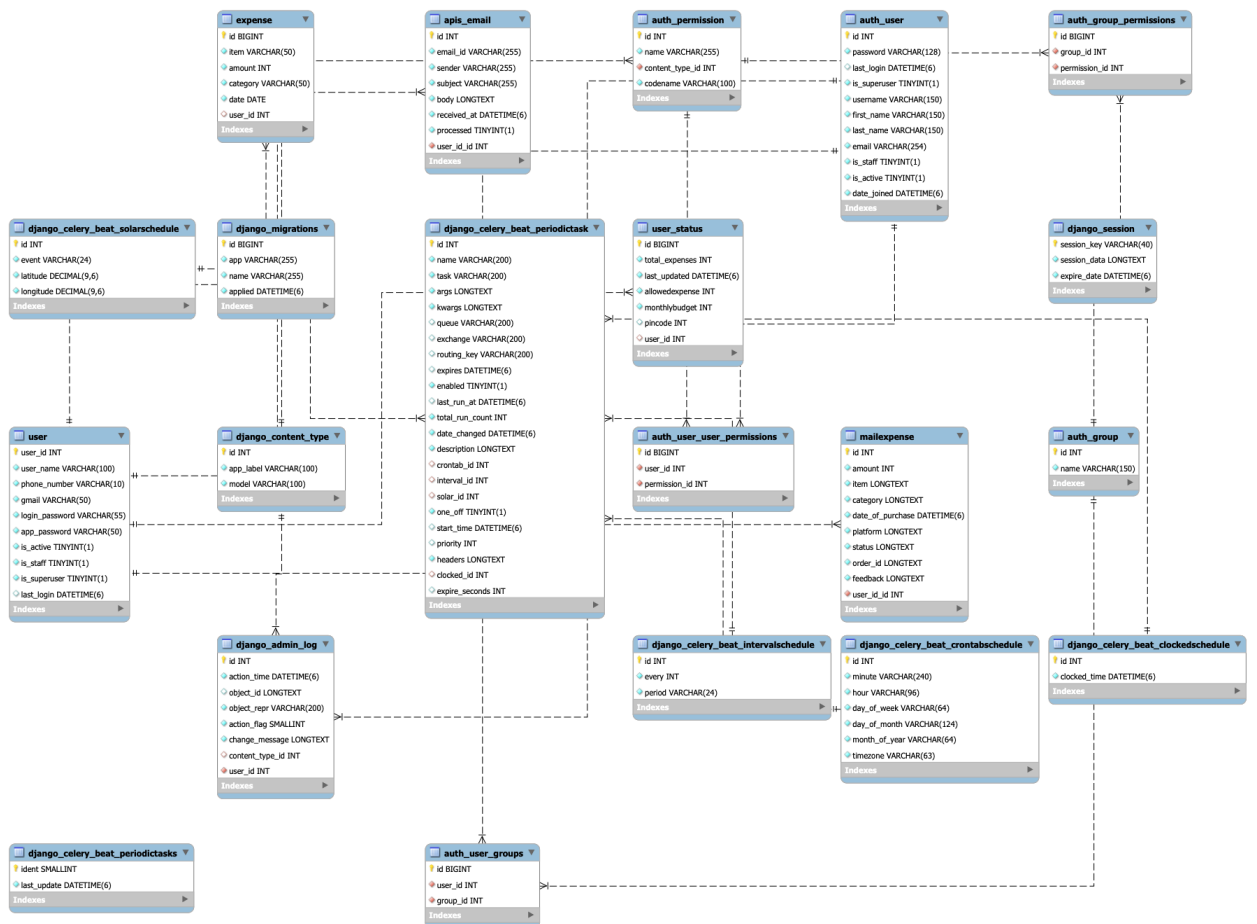
By leveraging these concepts, technologies and frameworks, the application effectively managed user data, provided real-time insights, and offered a seamless user experience across multiple platforms, including web browsers and chat interfaces.

Functionality Details

USER FLOW:



ER - DIAGRAMS :



LOGIN PAGE:

User Signup

Name:

Phone Number:

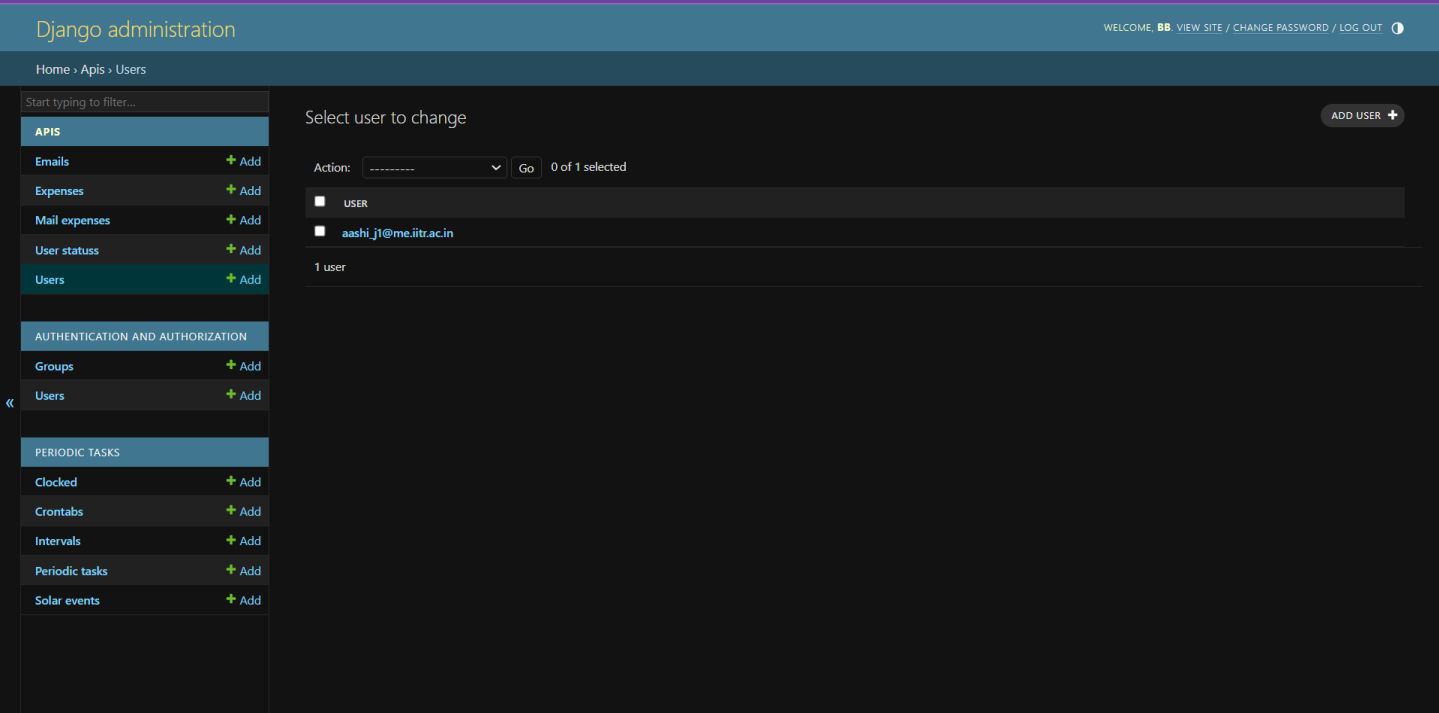
Gmail:

Login Password:

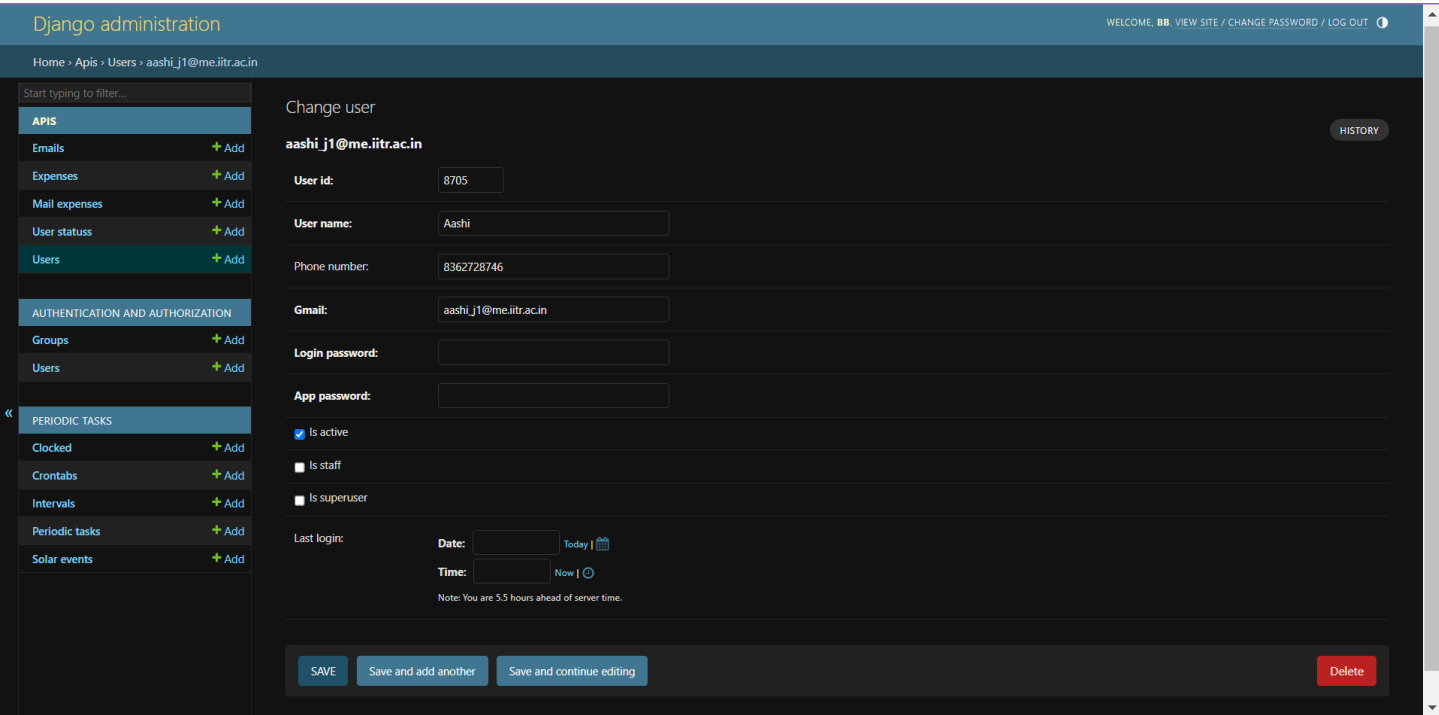
App Password:

[\(/login/\)>Sign Up](/login/)

SAMPLE DATABASE :



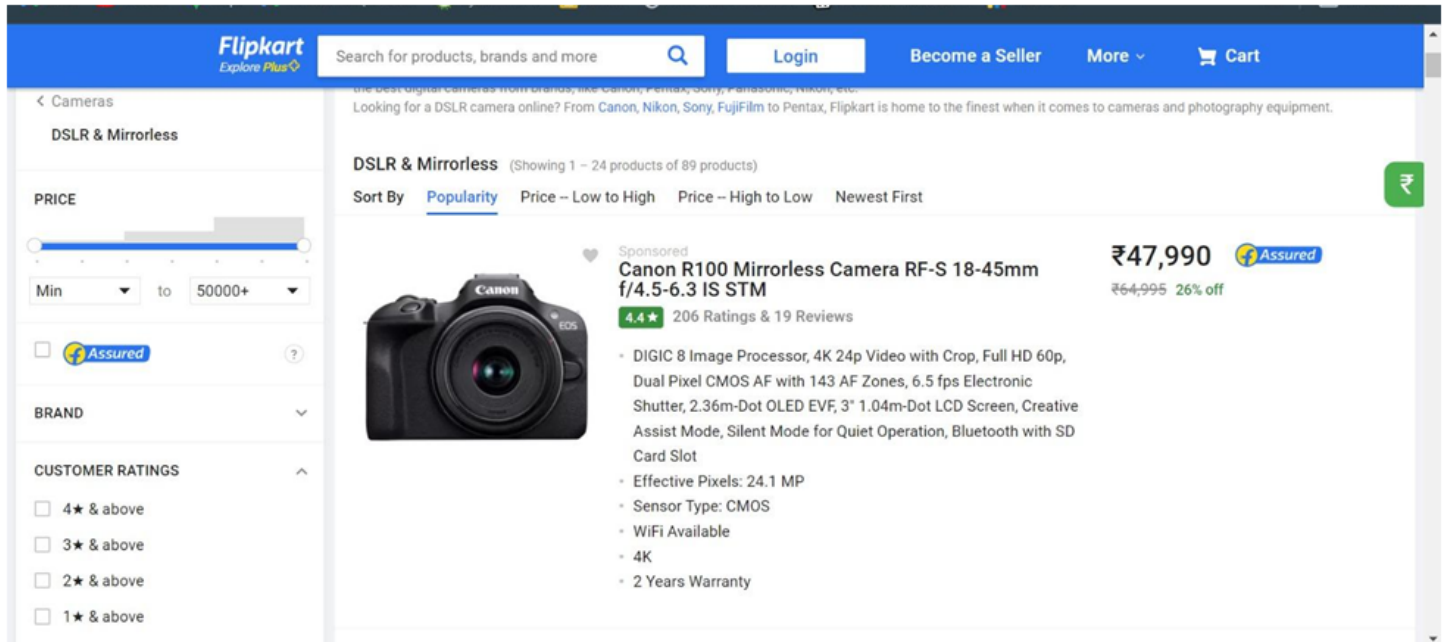
- This database stores the record of all the data provided by the user



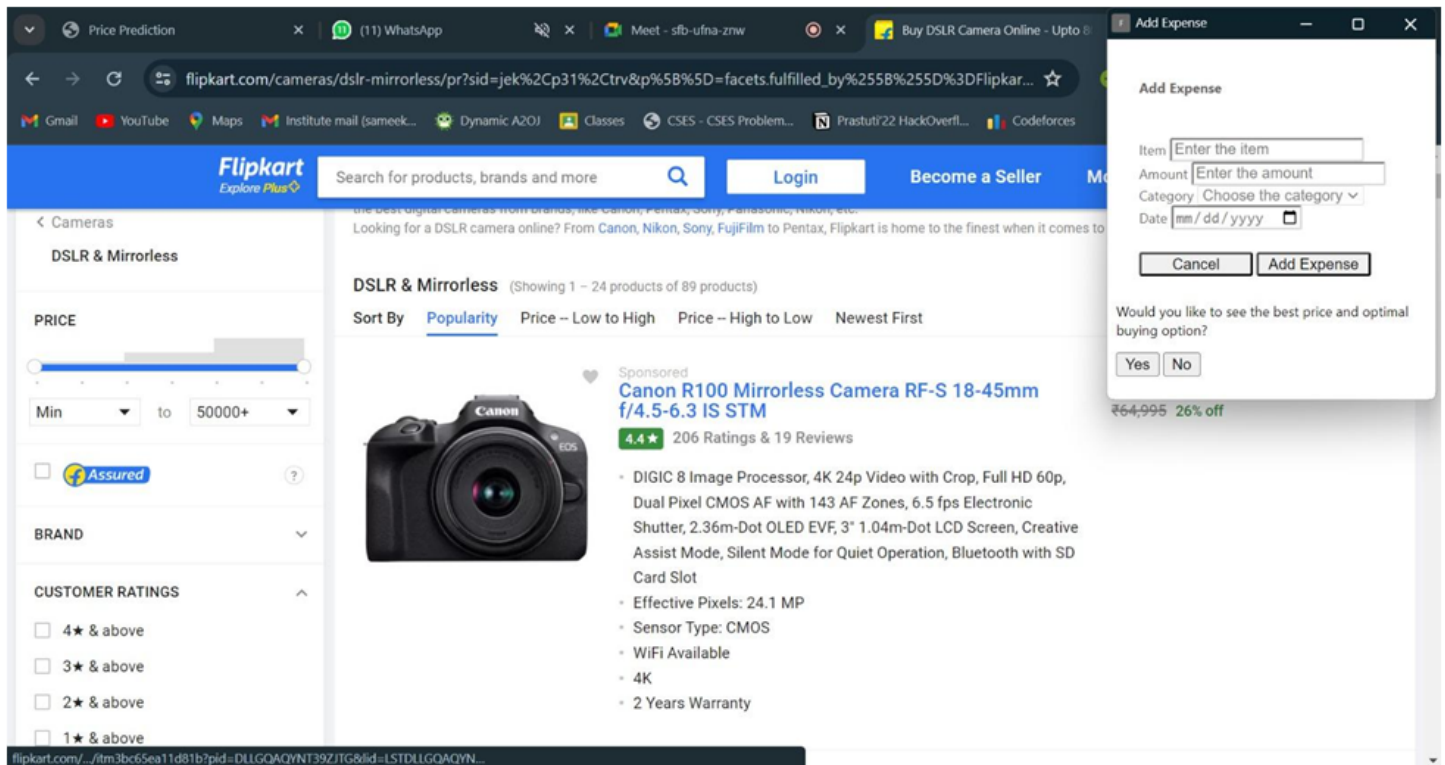
Features Implementation

EXTENSION:

- The green rupees icon indicates the Chrome Extension pop-up here :



- This indicates the Chrome Extension Pop-up in the top right window



- This shows the window for the expenses summary and the Chatbot to assist the user

Item	Amount	Category	Date	Actions
Coffee	300	Food	March 23, 2024	 
Vnrjv	2000	Travel	March 13, 2024	 
Clothes	2000	Clothes	March 21, 2024	 
Atomic habits	300	Stationery	March 21, 2024	 
Registers	3000	Other	March 23, 2024	 
Coffee	300	Other	March 23, 2024	 
Registers	3000	Other	March 23, 2024	 
Coffee	300	Other	March 23, 2024	 
Registers	3000	Other	March 23, 2024	 
Coffee	300	Other	March 23, 2024	 
Registers	3000	Other	March 23, 2024	 
Coffee	300	Other	March 23, 2024	 

CHATBOT:

- Chatbot conversation about our expense history.

Chatbot

ChatBot: Hello! I am DE Shaw Bot, a financial bot here to assist you with your expenses and provide personalized responses and advice based on your spending patterns. If you have any financial queries or need assistance with managing your expenses, feel free to ask!

You: please share how far i am from my goal

ChatBot: Based on the information you

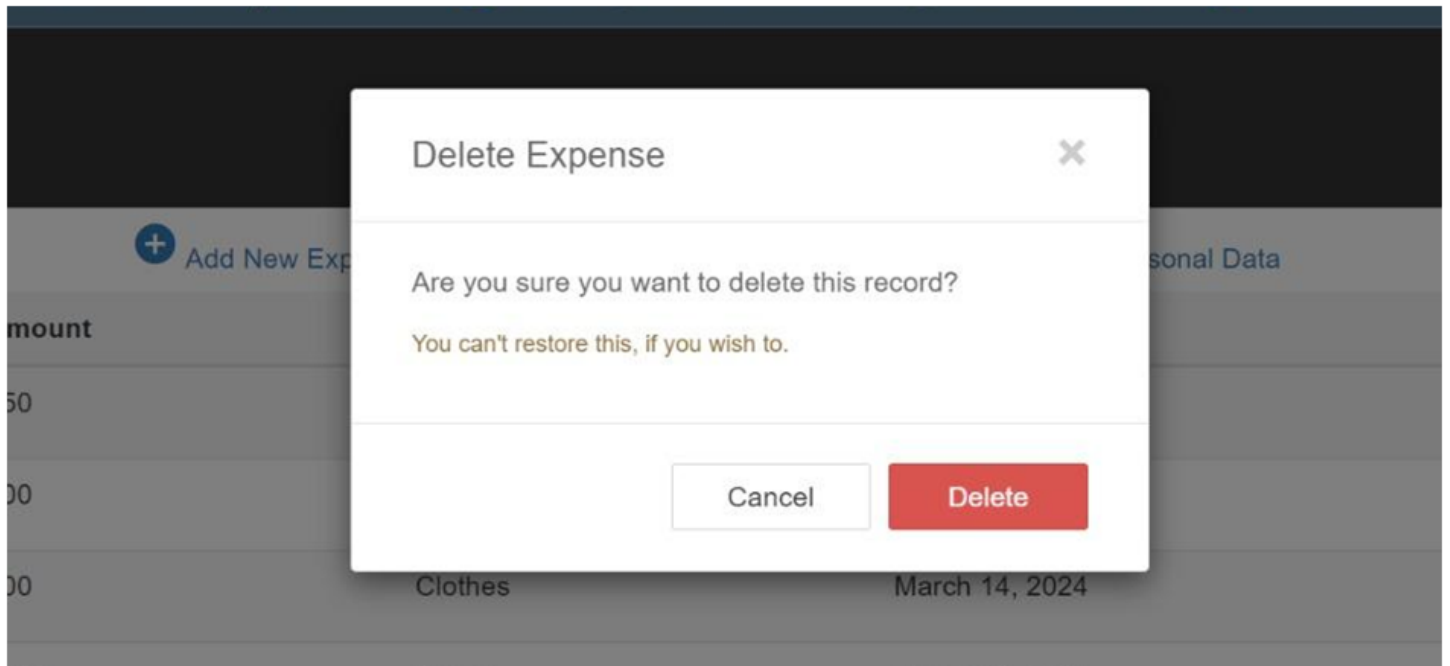
Chatbot

ChatBot: Hello! How can I assist you today? Feel free to share your expenses or ask any financial questions you may have.

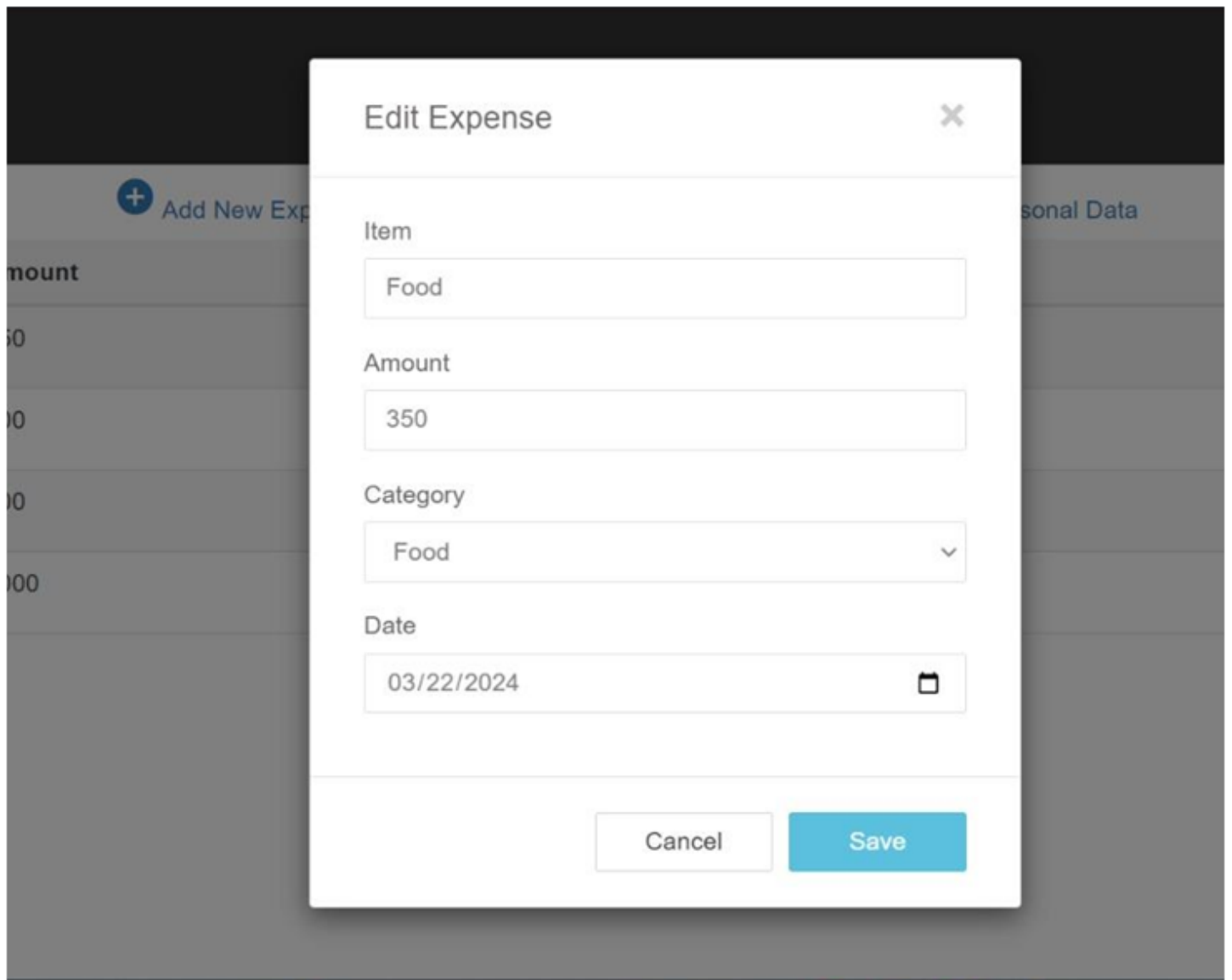
You: how am i doing in terms of expenses

ChatBot: Based on the expenses you have shared, here is a summary of your financial status: - Total Expenses: ₹6350 - Allowed Expense Limit: ₹4000 - Monthly Budget: ₹6000 You have

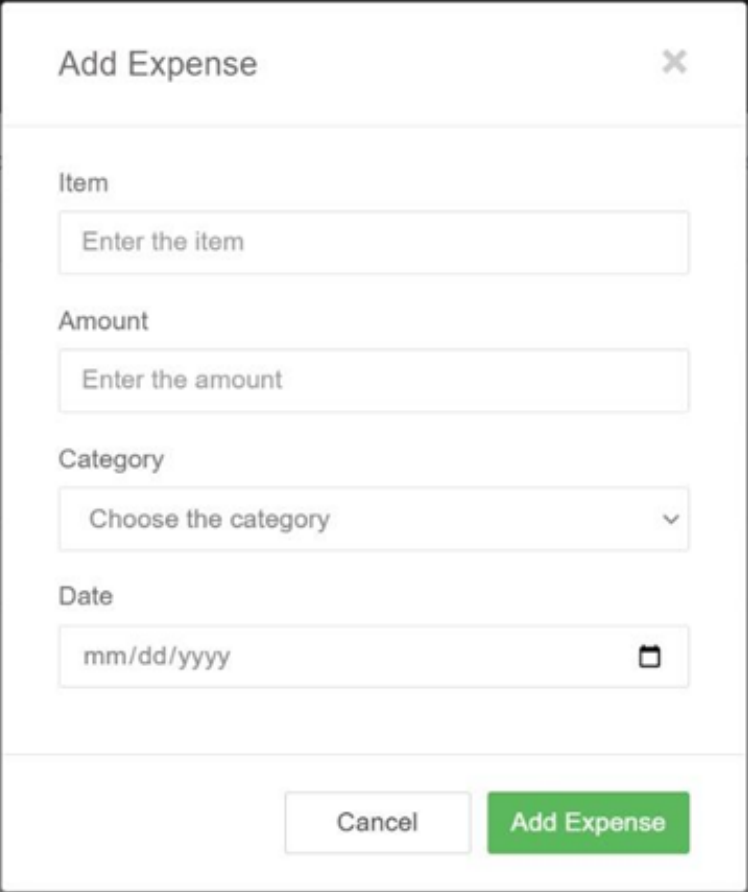
- Here we can delete an expense already added previously



- We can even edit our expenses incase any updating/editing is required



- Here, we can add new expenses with the details of our expenditure



Add Expense [Close]

Item

Amount

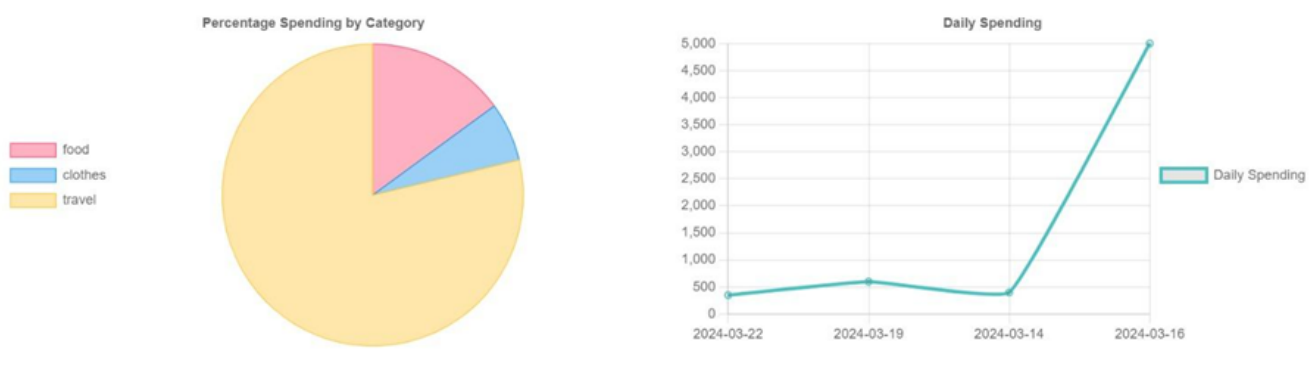
Category

Date

[Cancel] [Add Expense]

- We have represented our expenditures pictorially to improve user experience

runcode: 152140



- The user is prompted on the progress of the expenditures till now

User Status

Closer to Your Limit

Total Expenses: 2950

Last Updated: March 22, 2024, 3:42 p.m.

Allowed Expense: 4000

Monthly Budget: 6000

Current Balance: 3050

Score: 26.25

Pincode: 132140

User Status

Exceeded Your Limit

Total Expenses: 11950

Last Updated: March 22, 2024, 4:48 p.m.

Allowed Expense: 4000

Monthly Budget: 6000

Current Balance: -5950

Score: -198.75

Pincode: 132140

User Status

Good Going

Total Expenses: 2350

Last Updated: March 22, 2024, 4:49 p.m.

Allowed Expense: 4000

Monthly Budget: 6000

Current Balance: 3650

Score: 41.25

Pincode: 132140

FUTURE PREDICTION OF PRICES-OPTIMAL BUYING SOLUTION

- This shows the predicted price of the product in the future

Price Prediction

Product Name:
apple macbook air

Date (YYYY-MM-DD):
2023-05-22

Predict Price

[95278.16586987]

Maximum purchases month : 2023-01

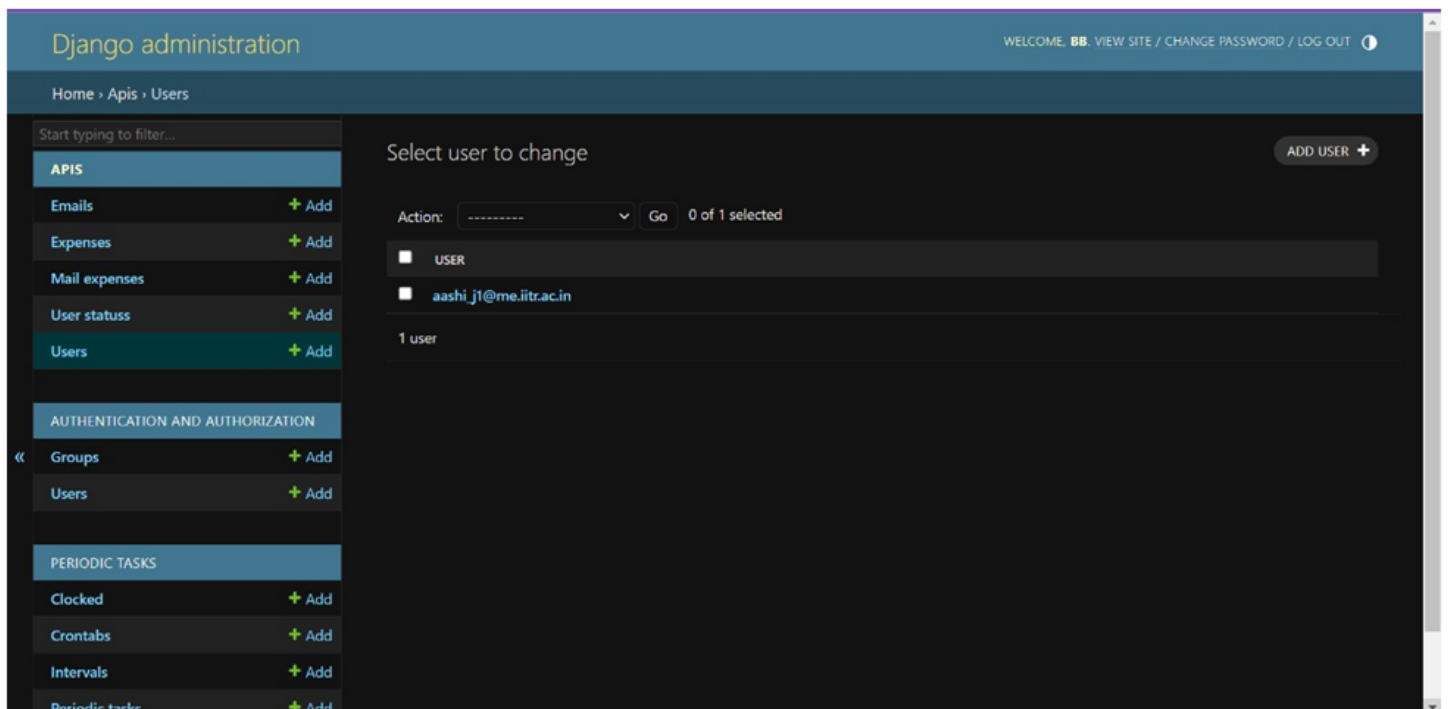
Lowest Price month : 2023-10

Price Comparison Result

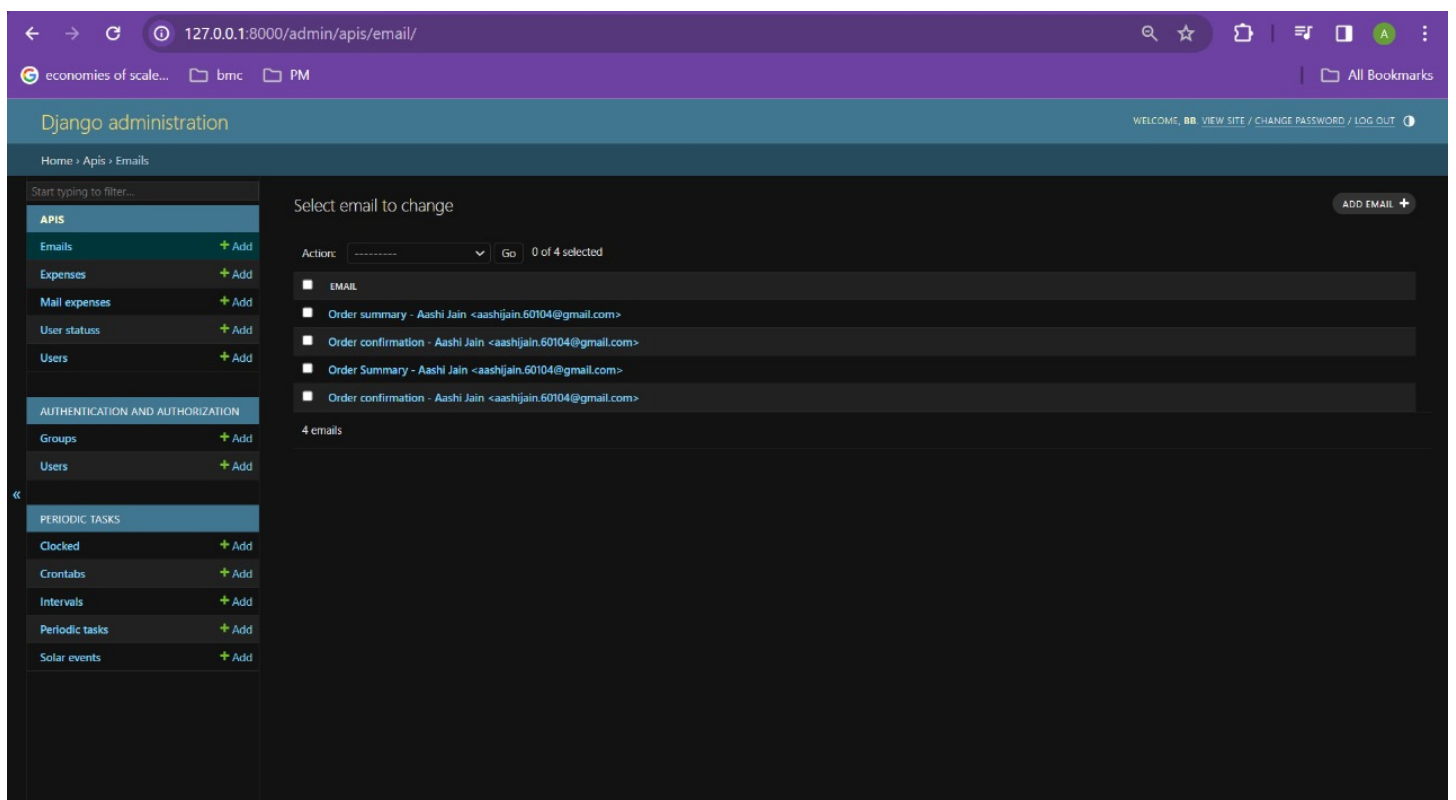
Price on Amazon is not available. Price on Flipkart is ₹81,490

EMAIL-INTEGRATION FOR AUTOMATED TRACKING

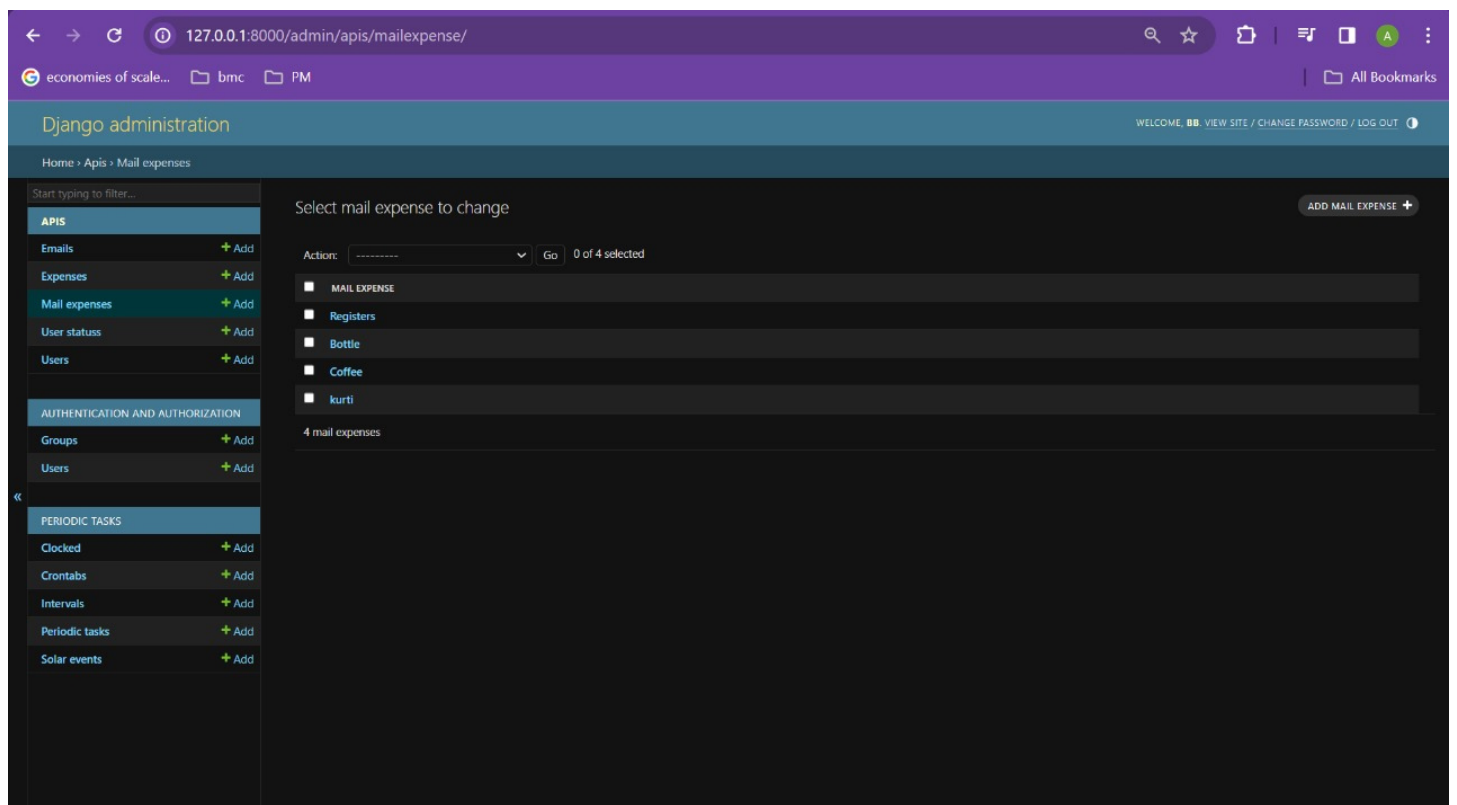
- In the case of Email Integration, this shows the addition of a user and the creation of user_id



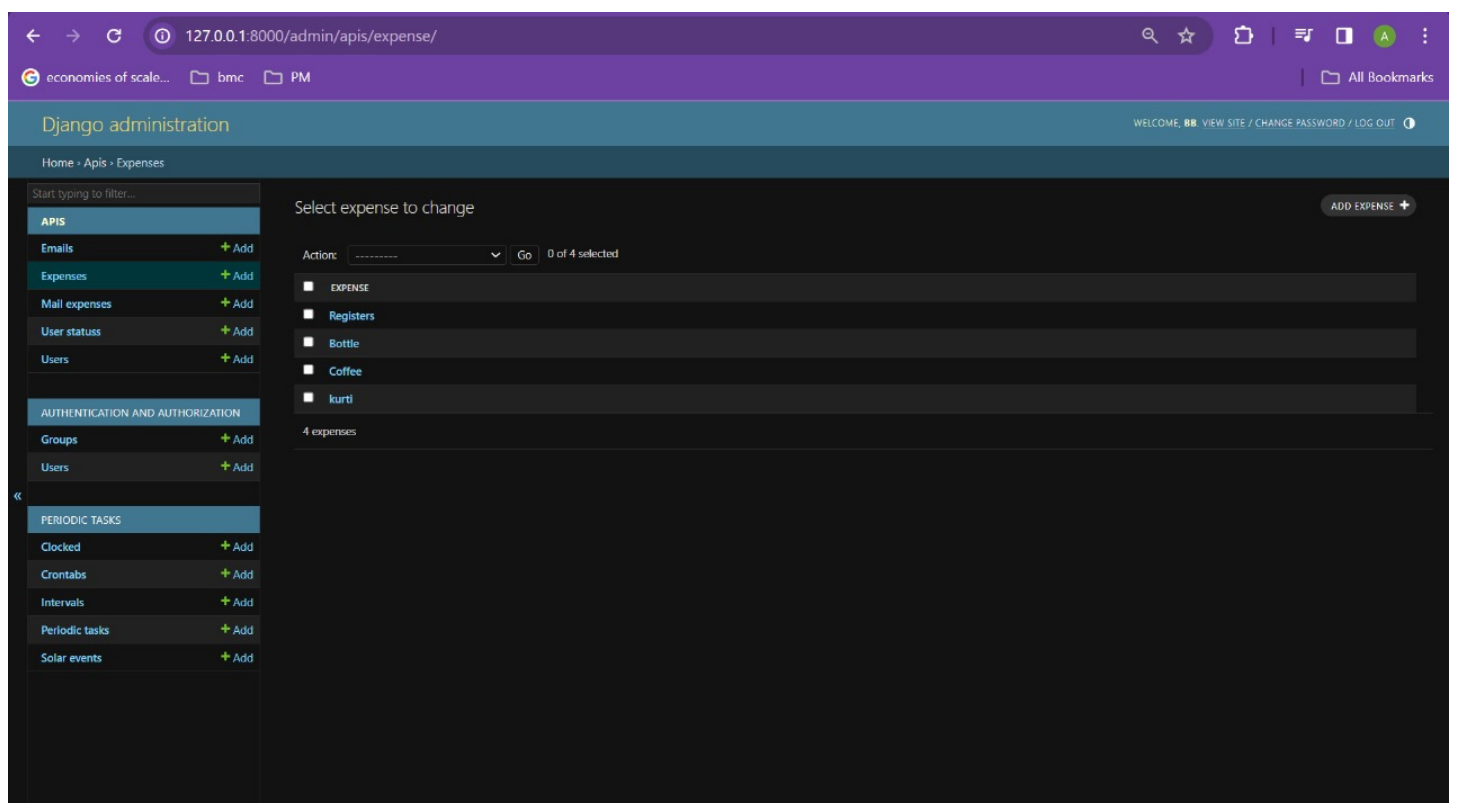
- Filtering of user's mail in inbox and only storing order-related/ecommerce- emails.



- Processing these emails using Gemini to get expenses in a standardized format



- Addition of all the Expenses tracked from mail into a common Expenses Table



- The final/updated Home Page for the given user

The chart, titled "Daily Spending", displays a single data point for the date 2024-03-23. The y-axis represents the spending amount, ranging from 8,799.0 to 8,801.0 in increments of 0.2. The data point is marked with a teal circle at the value 8,800.0. A legend on the right indicates that the teal line represents "Daily Spending".

Date	Daily Spending
2024-03-23	8,800.0

Future Scope

1) Integration Of Google Maps for making a friends community :

- Utilizing Google Maps API, Zenfi can allow users to create groups with their friends or acquaintances based on geographical proximity. Users can invite others nearby to join their budgeting group, forming a social network centered around financial goals and accountability.
- The integration could include features such as location-based friend suggestions, where users are prompted to connect with individuals in their vicinity who share similar financial interests or goals.

2) Incorporate a leaderboard and ranking system :

- Implement personalized leaderboards tailored to users' specific goals or interests, allowing individuals to compare their progress against peers with similar financial profiles or demographics.
- Establish a rewards program within Zenfi. Provide tangible benefits such as cash-back bonuses, gift cards, or exclusive discounts from partner merchants, encouraging users to actively participate in budgeting activities and strive for higher rankings.

3) Google Oauth2 for Gmail-Integration :

- Instead of asking the user for their app password, we can utilize the Google OAuth2 key to access and read the user's inbox. This key will be issued to our project during verification when it reaches production level.

